

IDENTIFICATION OF MYCOBACTERIUM SPECIES THROUGH BIOINFORMATICS ANALYSIS

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ABSTRACT

Mycobacterium species are rod-shaped, acid-fast, Gram-positive organisms within the Actinobacteria phylum. Many species, including *M. tuberculosis* are well-known human pathogens. Given the diversity present within mycobacterial species and their clinical significance, there is a need for species identification and delineation using robust methods. Bioinformatics analysis provides insights into the species’ genomics, phylogeny, and biology. Genome assembly and annotation were crucial in the present analysis for understanding the genetic content and functionality, enabling downstream analyses. *Mycobacterium* species identification is primarily based on 16S rRNA and core genome analysis. Phylogenetic tree analysis and in silico DNA hybridisation were used to confirm the identification of *M. tuberculosis* using unknown sequencing reads from the dataset “Data 03”.

INTRODUCTION

Mycobacterium genus:

- actinobacterium phylum (Gao and Gupta 2012)
- acid-fast, rod-shaped, Gram-positive
- pathogenic *M. tuberculosis* and *M. leprae* (Medjahed *et al.* 2010)

Bioinformatics analysis, including phylogenetic tree analysis and in silico DNA hybridisation, were used to confirm the identity of the unknown *Mycobacterium* species.

RESULTS AND DISCUSSION

Phylogenetic tree analysis

- *M. tuberculosis*, *M. kansasii*, and *M. absense* were similarly grouped as in another study by Gupta *et al.* (2018)
- Data 03 and *M. tuberculosis* were observed to be closely related species in both the 16S rRNA- and Core genome SNP-based trees (Figure 1)

In silico DNA hybridization

- ANIb value of 95% is used as a species cutoff (Varghese *et al.* 2015). DDH value of 70% was used to define species boundary, and 80% to define subspecies boundary (Espariz *et al.* 2016).
- Data 03 and *M. tuberculosis* had ANIb values of 99.86-99.91% and DDH value of 98.6%, and were defined as the same species (Figure 2)

The possible identity of Data 03 is *Mycobacterium tuberculosis*

OBJECTIVES

- To identify the *Mycobacterium* species using unknown sequencing reads
- To employ a comprehensive bioinformatics analysis, including quality control, genome assembly, genome annotation, and phylogenetic tree analysis.

METHODOLOGY

“Data 03” forward and reverse reads were sequenced using Illumina Hiseq 1500

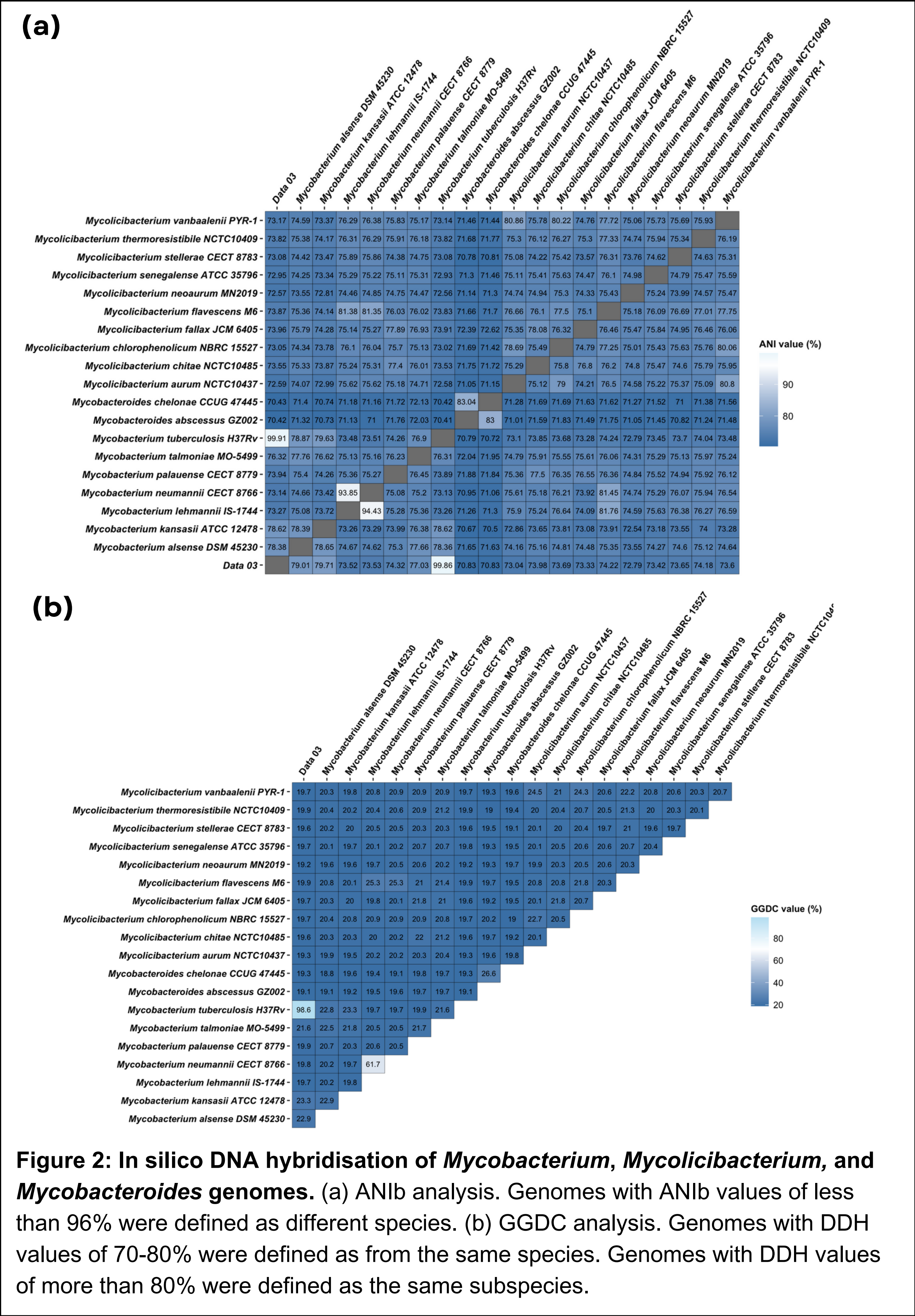
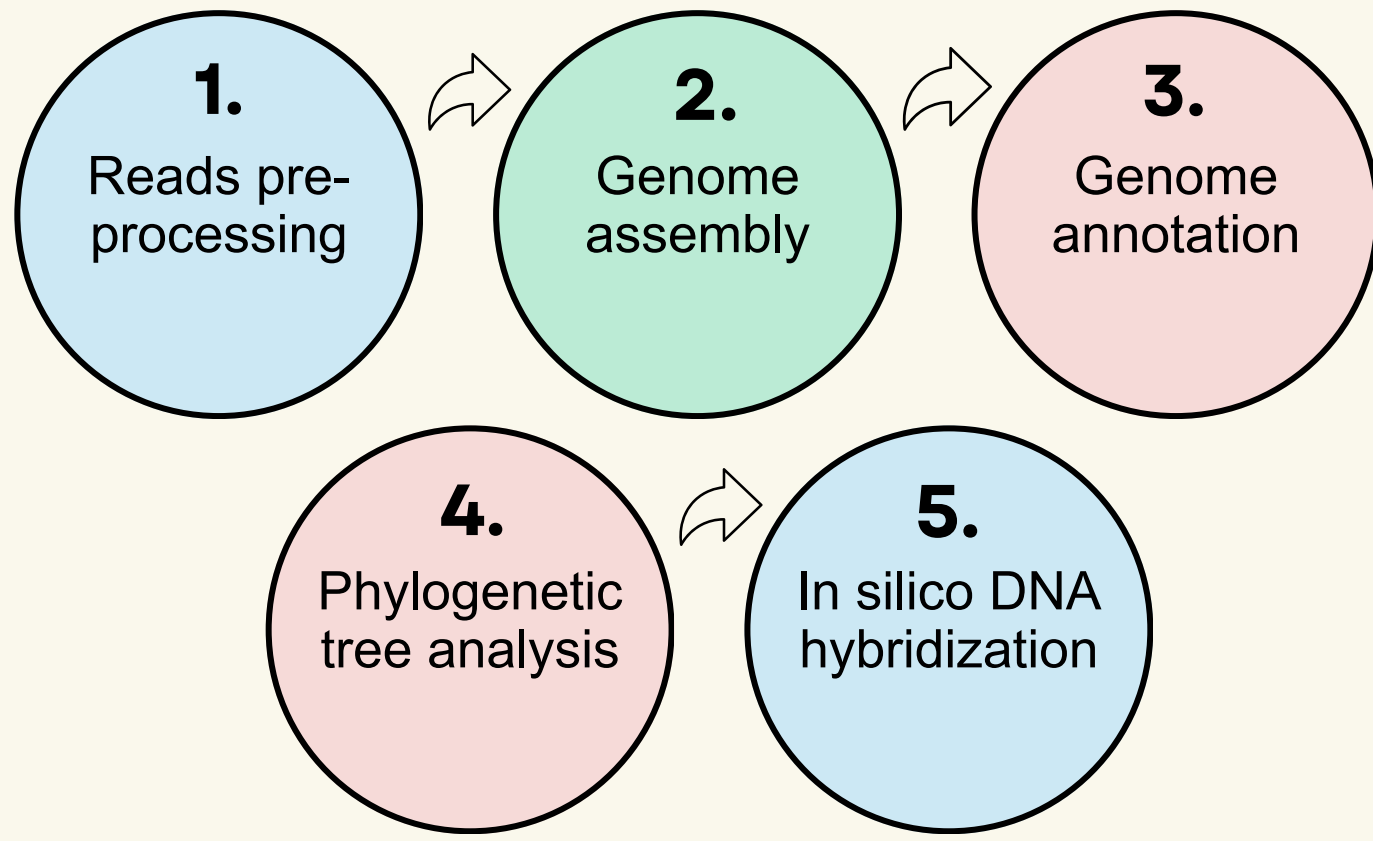


Figure 2: In silico DNA hybridisation of *Mycobacterium*, *Mycolicibacterium*, and *Mycobacteroides* genomes. (a) ANIb analysis. Genomes with ANIb values of less than 96% were defined as different species. (b) GGDC analysis. Genomes with DDH values of 70-80% were defined as from the same species. Genomes with DDH values of more than 80% were defined as the same subspecies.

CONCLUSION

- Phylogenetic tree analysis and in silico DNA hybridisation confirmed the identity of Data 03 to be *Mycobacterium tuberculosis*
- Further study: Antibiotic resistance genes and/or virulent factor prediction, multilocus sequence typing (Kayani *et al.* 2018)

References:

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